

# Rates-Motion-Time-and-Geometry — ASCII Master

|   |                                                                   |  |
|---|-------------------------------------------------------------------|--|
| + | PANEL 01 – RATE MODEL                                             |  |
| + | Quantity equals rate multiplied by time.                          |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 02 – TIME AND WORK                                          |  |
| + | Add work rates, not completion times.                             |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 03 – PIPES AND LEAKS                                        |  |
| + | Treat filling positive and emptying negative.                     |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 04 – SPEED AND DISTANCE                                     |  |
| + | Match units before using distance equals speed times time.        |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 05 – RELATIVE MOTION                                        |  |
| + | Add opposite-direction speeds and subtract same-direction speeds. |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 06 – TRAINS AND BOATS                                       |  |
| + | Include object length and current speed correctly.                |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 07 – RACES AND TRACKS                                       |  |
| + | Use relative distance on linear or circular paths.                |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 08 – CLOCKS                                                 |  |
| + | Use the 5.5-degree-per-minute relative hand speed.                |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 09 – CALENDARS                                              |  |
| + | Reduce day shifts modulo seven and handle leap-year rules.        |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 10 – GEOMETRY                                               |  |
| + | Use properties before coordinates or formulas.                    |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 11 – MENSURATION                                            |  |
| + | Distinguish length, area and volume units.                        |  |
|   |                                                                   |  |
|   | v                                                                 |  |
| + | PANEL 12 – VERIFICATION                                           |  |
| + | Check dimensions, magnitude, endpoints and physical possibility.  |  |
|   |                                                                   |  |